NETWORKING

TLS Handshake Visual Guide

How HTTPS builds a private tunnel
on a public internet — in ~2 round trips.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Goal: Private + Authenticated

Three guarantees before any application data flows.

Confidentiality

Nobody can read the traffic (encryption).

Integrity

Nobody can tamper with it undetected (MAC/AEAD).

Authentication

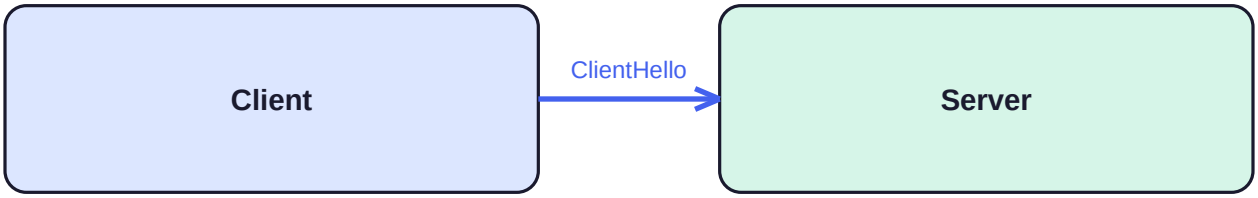
You really are talking to example.com (certificates).

TLS sits between TCP and HTTP. HTTPS = HTTP inside a TLS tunnel.

STEP 2/6

Step 1 — ClientHello

Client offers what it supports.

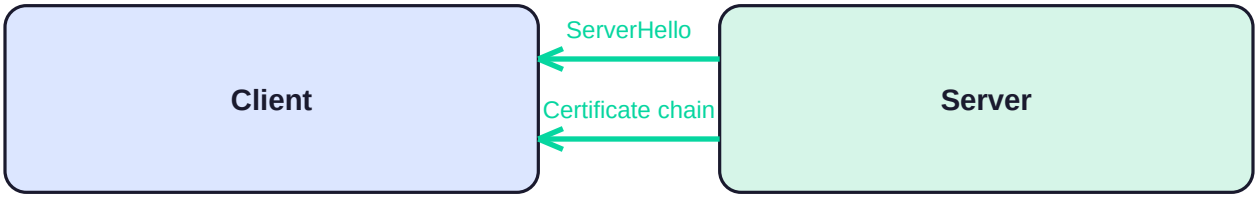


- TLS versions it speaks (1.2, 1.3)
- Cipher suites, ranked by preference
- Random bytes (anti-replay)
- SNI: the hostname it wants (for virtual hosting)

STEP 3/6

Step 2 — ServerHello + Certificate

Server picks parameters and proves its identity.



Certificate contains: domain, public key, issuer signature, expiry.

Client verifies: signature chains to a trusted CA + name matches + not expired.

Chain of trust

leaf cert → intermediate CA → root CA (in OS/browser store)

STEP 4/6

Step 3 — Key Exchange

Agree on session keys without ever sending them.

TLS 1.3: Ephemeral Diffie-Hellman (ECDHE)

- Each side generates a one-time (ephemeral) key pair and sends the public half.
- Both combine their private half with the other's public half → same shared secret.
- An eavesdropper sees only public values — deriving the secret requires solving discrete log.
- Ephemeral = forward secrecy: stealing the server's long-term key later can't decrypt past sessions.

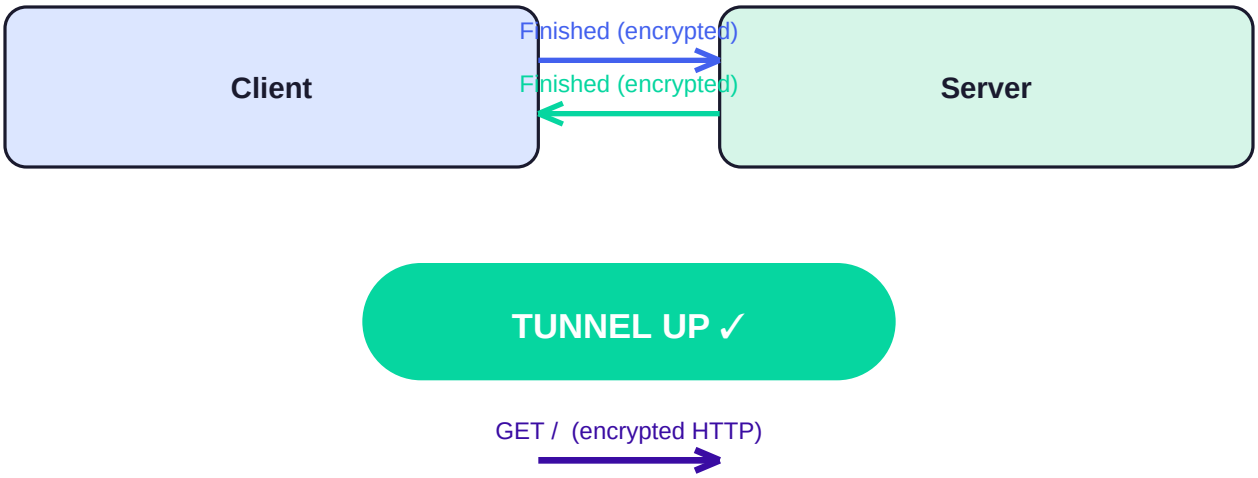
Result: symmetric session keys

AES-GCM or ChaCha20-Poly1305 — fast bulk encryption + integrity

STEP 5/6

Steps 4–5 — Finished, Then Encrypted Data

Both sides confirm, then the tunnel carries HTTP.



From here on, everything — headers, cookies, body — is encrypted.

STEP 6/6

TLS 1.3 vs 1.2 + Session Resumption

Faster handshakes for returning visitors.

Handshake RTTs

TLS 1.2: 2 RTT → TLS 1.3: 1 RTT

0-RTT resumption

Repeat visitors can send data with the first flight (replay risk — use carefully).

Removed

1.3 dropped RSA key exchange, CBC modes, renegotiation — smaller, safer.

Certificates

Still X.509; automation via ACME/Let's Encrypt made HTTPS ubiquitous.

STEP ✓I✓

Key Takeaways

TLS Handshake — Remember

- ClientHello offers parameters; ServerHello + certificate picks them and proves identity.
- Certificates chain to trusted roots; the client checks signature, name, and expiry.
- ECDHE key exchange creates session keys no eavesdropper can derive — with forward secrecy.
- Finished messages confirm; then HTTP flows inside the encrypted tunnel.
- TLS 1.3: 1-RTT handshake, 0-RTT resumption, and a trimmed, safer cipher list.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev